Cambridge (CIE) A Level Maths: Pure 3

Your notes

Vectors in 3 Dimensions

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- Problem Solving using 3D Vectors

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Your notes

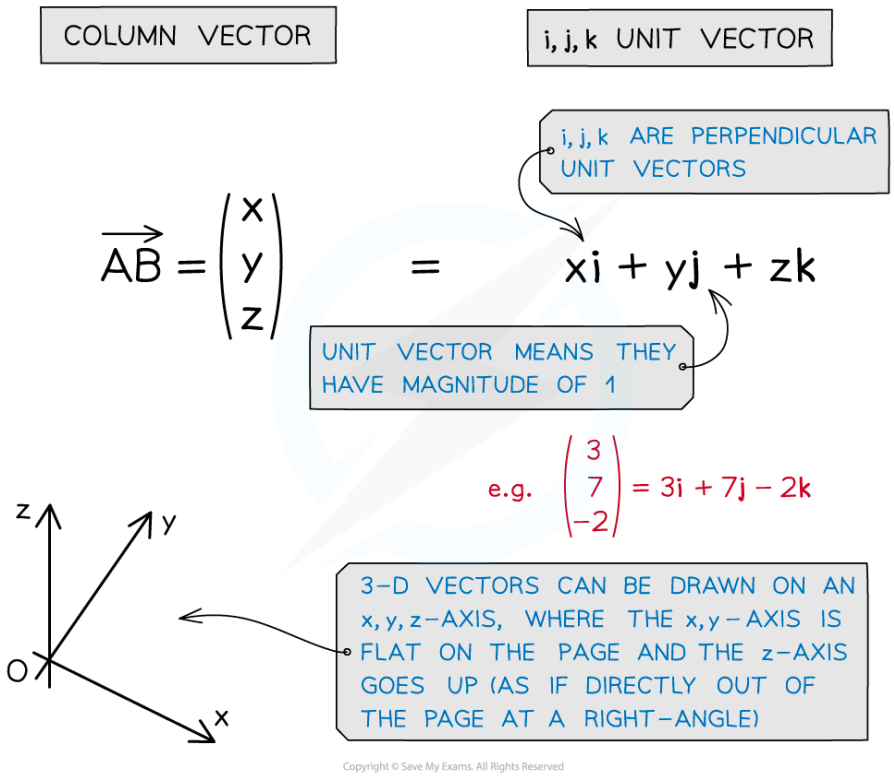
Vectors in 3 dimensions

What is a 3D vector?

- Vectors represent a movement of a certain **magnitude** (size) in a given **direction**

  You should have already come across (2D) **vectors** at AS (see Basic Vectors)

- 3D vectors describe the **position of a point** in a 3D space in relation to the origin
- They can be represented in different ways such as a **column vector** or in **i, j, k unit vector form**



Magnitude of a 3D vector

- The magnitude of a 3D vector is simply its size
- Like 2D vectors we can find the **magnitude** using Pythagoras' theorem (see Magnitude Direction)

Your notes

MAGNITUDE

$$|\mathbf{a}| = |x\mathbf{i} + y\mathbf{j} + z\mathbf{k}| = \left|\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right| = \sqrt{x^2 + y^2 + z^2}$$

REMEMBER THERE ARE LOTS OF DIFFERENT WAYS TO REPRESENT THE SAME VECTOR

$$|\mathbf{a}| = |\overrightarrow{AB}| = \left|\begin{pmatrix} 3 \\ 7 \\ -2 \end{pmatrix}\right| = |3\mathbf{i} + 7\mathbf{j} - 2\mathbf{k}|$$

A VECTOR'S MAGNITUDE IS SOMETIMES REFERRED TO AS ITS MODULUS

$$|\overrightarrow{AB}| = \sqrt{3^2 + 7^2 + 2^2} = \sqrt{62} = 7.874... = 7.9 \text{ (1 dp)}$$

YOU CAN IGNORE MINUS SIGN

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- For 3D **position vectors** we can find the distance between two points
- By using the respective co-ordinates we can calculate the magnitude of the vector between them:

DISTANCE BETWEEN TWO POINTS FROM POSITION VECTORS

(x_1, y_1, z_1) AND $(x_2, y_2, z_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

WHERE x, y, z ARE FROM POSITION VECTORS $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

e.g. FIND $|\overrightarrow{AB}|$ FOR POSITION VECTORS A AND B, $5\mathbf{i} + 3\mathbf{j} - 7\mathbf{k}$ AND $3\mathbf{i} - 6\mathbf{j} + 5\mathbf{k}$ RESPECTIVELY

A HAS CO-ORDINATES $(5, 3, -7)$

B HAS CO-ORDINATES $(3, -6, 5)$

$$|\overrightarrow{AB}| = \sqrt{(5-3)^2 + (3 - -6)^2 + (-7-5)^2} = \sqrt{4 + 81 + 144}$$

$$= \sqrt{229} = 15.1 \text{ UNITS (1 dp)}$$

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3D vector addition, scalars, parallel vectors and unit vectors

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- 3D vectors work in the same way as 2D vectors, just in three dimensions rather than two
- **Vector addition** and subtraction and scalar multiplication can be carried out in exactly the same way, this time involving **i, j** and **k** or x, y and z
- 3D vectors are also **parallel** if one is a **multiple** of the other

COLUMN VECTOR

$$\mathbf{a} = \begin{pmatrix} 2 \\ 6 \\ -4 \end{pmatrix} \qquad \mathbf{b} = \begin{pmatrix} -3 \\ 7 \\ 4 \end{pmatrix}$$

$$2\mathbf{a} - 3\mathbf{b} = 2 \times \begin{pmatrix} 2 \\ 6 \\ -4 \end{pmatrix} - 3 \times \begin{pmatrix} -3 \\ 7 \\ 4 \end{pmatrix}$$

$$= \begin{pmatrix} 4 \\ 12 \\ -8 \end{pmatrix} - \begin{pmatrix} -9 \\ 21 \\ 12 \end{pmatrix} = \begin{pmatrix} 13 \\ -9 \\ -20 \end{pmatrix}$$

i, j, k NOTATION

$$\mathbf{a} = -3\mathbf{i} + 9\mathbf{j} + 2\mathbf{k}$$

$$\mathbf{b} = 8\mathbf{i} - 6\mathbf{j} + 4\mathbf{k}$$

$$2\mathbf{a} + \frac{1}{2}\mathbf{b} = 2(-3\mathbf{i} + 9\mathbf{j} + 2\mathbf{k}) + \frac{1}{2}(8\mathbf{i} - 6\mathbf{j} + 4\mathbf{k})$$

$$= (-6\mathbf{i} + 18\mathbf{j} + 4\mathbf{k}) + (4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k})$$

$$= -2\mathbf{i} + 15\mathbf{j} + 6\mathbf{k}$$

3-D VECTORS

PARALLEL VECTORS

$$\mathbf{a} = \begin{pmatrix} 9 \\ -6 \\ 3 \end{pmatrix} \qquad \mathbf{b} = \begin{pmatrix} 6 \\ -4 \\ 2 \end{pmatrix}$$

a IS PARALLEL TO **b** AS

$$\mathbf{b} = \frac{2}{3}\mathbf{a} \qquad \frac{2}{3}\begin{pmatrix} 9 \\ -6 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \\ 2 \end{pmatrix}$$

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- **Unit vectors** in 3D are found in exactly the same way as in 2D

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TO FIND THE **UNIT VECTOR** IN THE SAME DIRECTION AS A GIVEN VECTOR, SIMPLY DIVIDE THE VECTOR BY ITS MAGNITUDE

- THE UNIT VECTOR IN THE SAME DIRECTION AS VECTOR $\overrightarrow{AB}$ IS $\dfrac{\overrightarrow{AB}}{|\overrightarrow{AB}|}$
- THE UNIT VECTOR IN THE SAME DIRECTION AS VECTOR $\mathbf{a}$ IS $\dfrac{\mathbf{a}}{|\mathbf{a}|}$

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e.g. FIND THE UNIT VECTOR IN THE SAME DIRECTION AS THE VECTOR $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$

$$|\mathbf{a}| = \sqrt{3^2 + (-4)^2 + 5^2} = 5\sqrt{2}$$

$$\text{UNIT VECTOR} = \frac{\mathbf{a}}{|\mathbf{a}|} = \frac{3\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}}{5\sqrt{2}} = \frac{1}{5\sqrt{2}}(3\mathbf{i} - 4\mathbf{j} + 5\mathbf{k})$$

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Worked Example

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Points A and B have position vectors $-\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$ and $3\mathbf{i} + 7\mathbf{j} - 2\mathbf{k}$ respectively.

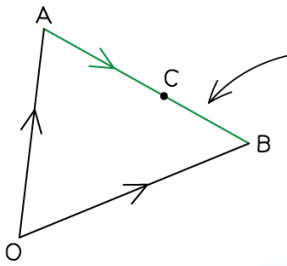
Point C divides the line AB in the ratio 3:2.

a) Find the position vector of C.

b) Calculate the distance of C from the origin.

Give your answer to 2 decimal places.

a)



START BY DRAWING A DIAGRAM TO HELP, DON'T WORRY ABOUT THE 3-D ASPECT, THE VECTORS WILL TAKE CARE OF THAT!

FIND VECTORS FOR ANY UNKNOWN PARTS OF THE DIAGRAM

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (3\mathbf{i} + 7\mathbf{j} - 2\mathbf{k}) - (-\mathbf{i} - 2\mathbf{j} + 5\mathbf{k})$$
$$= (3 - -1)\mathbf{i} + (7 - -2)\mathbf{j} + (-2 - 5)\mathbf{k}$$
$$= 4\mathbf{i} + 9\mathbf{j} - 7\mathbf{k}$$

$$\overrightarrow{AC} = \frac{3}{5}(4\mathbf{i} + 9\mathbf{j} - 7\mathbf{k})$$
$$= \frac{12}{5}\mathbf{i} + \frac{27}{5}\mathbf{j} - \frac{21}{5}\mathbf{k}$$

RATIO 3:2 MEANS C IS $\frac{3}{5}$ ALONG $\overrightarrow{AB}$

$$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = (-\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}) + \left(\frac{12}{5}\mathbf{i} + \frac{27}{5}\mathbf{j} - \frac{21}{5}\mathbf{k}\right)$$
$$= \frac{7}{5}\mathbf{i} + \frac{17}{5}\mathbf{j} + \frac{4}{5}\mathbf{k}$$

POSITION VECTOR NEEDS TO START FROM ORIGIN

b) $$|\overrightarrow{OC}| = \left|\frac{7}{5}\mathbf{i} + \frac{17}{5}\mathbf{j} + \frac{4}{5}\mathbf{k}\right|$$

DISTANCE = MAGNITUDE

$$|\overrightarrow{OC}| = \sqrt{\left(\frac{7}{5}\right)^2 + \left(\frac{17}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \frac{\sqrt{354}}{5}$$

$= 3.76$ UNITS (2 dp)

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Problem solving using 3D vectors

How are 3D vectors used for problem solving?

- 3D vector problems can be solved using the same principles as **2D vector problems** (see Problem Solving using Vectors)
- Vectors can be used to prove two lines are **parallel**, to show points are **collinear** (lie on the same straight line) or to find missing vertices of a given shape

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VECTORS ARE PARALLEL IF ONE IS A SCALAR **MULTIPLE** OF THE OTHER

PARALLEL VECTORS

$$\mathbf{a} = \begin{pmatrix} 9 \\ -6 \\ 3 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 6 \\ -4 \\ 2 \end{pmatrix}$$

$\mathbf{a}$ IS PARALLEL TO $\mathbf{b}$ AS

$$\mathbf{b} = \frac{2}{3}\mathbf{a} \qquad \frac{2}{3}\begin{pmatrix} 9 \\ -6 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \\ 2 \end{pmatrix}$$

THREE POINTS P, Q, R ARE **COLLINEAR** IF $\overrightarrow{PQ}$ AND $\overrightarrow{PR}$ ARE PARALLEL

SHOW THAT POINTS A(1, 2, 3) B(3, 8, 1) AND C(7, 20, −3) ARE COLLINEAR

$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ 8 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \\ -2 \end{pmatrix}$$

$$\overrightarrow{AC} = \begin{pmatrix} 7 \\ 20 \\ -3 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 18 \\ -6 \end{pmatrix}$$

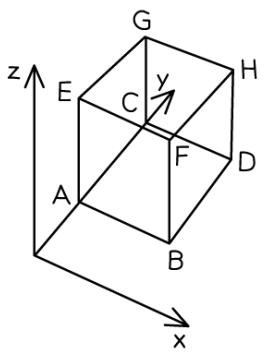
$\overrightarrow{AC} = 3\overrightarrow{AB}$ SO $\overrightarrow{AC}$ IS PARALLEL TO $\overrightarrow{AB}$.

∴ ABC ARE COLLINEAR

USE COORDINATES TO GIVE POSITION VECTORS THEN FIND AB AND AC TO PROVE THEY ARE COLLINEAR

3-D VECTOR PROBLEMS

PARALLEL SIDES OF A SHAPE SUCH AS A CUBOID HAVE THE **SAME VECTOR**



A CUBOID HAS POINTS B AND G SUCH THAT B HAS POSITION VECTOR $5\mathbf{i} + 3\mathbf{j}$ AND G HAS POSITION VECTOR $7\mathbf{j} + 6\mathbf{k}$.

FIND THE POSITION VECTOR OF POINT H IN UNIT VECTOR FORM

$$H = \begin{pmatrix} 5 \\ 7 \\ 6 \end{pmatrix} = 5\mathbf{i} + 7\mathbf{j} + 6\mathbf{k}$$

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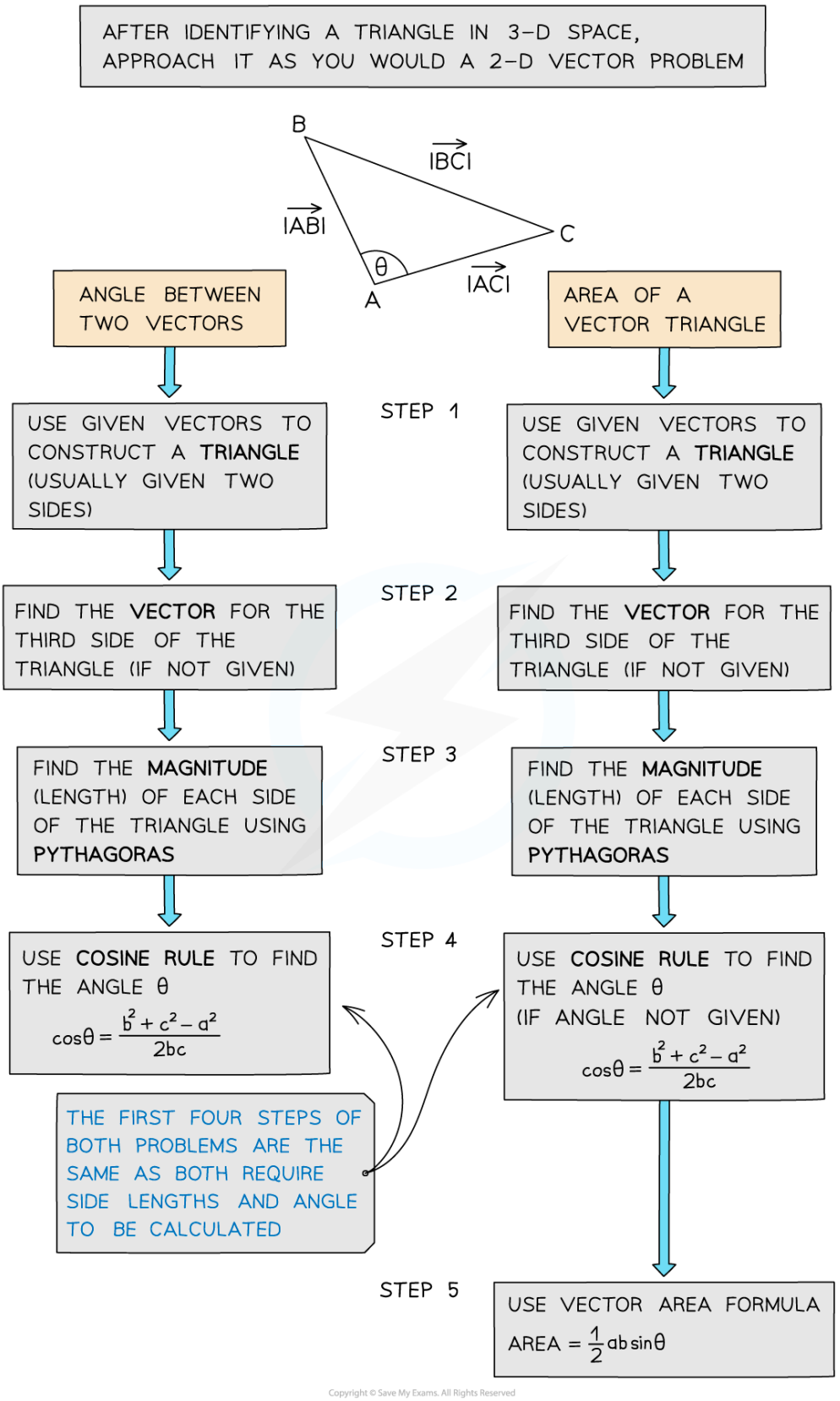
When do I use trigonometric in 3D vector problems?

- 3D vector problems can also involve using trigonometry to:
  - find the **angle between two vectors** using **Cosine Rule**
  - find the **area of a triangle** using a variation of **Area Formula**

Your notes



- You can find the angle between a vector and any one of the coordinate axes by using the following formulae

Your notes

IF θ_x, θ_y AND θ_z ARE RESPECTIVELY THE ANGLES THAT THE VECTOR $\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ MAKES WITH THE x–, y– AND z–AXES, THEN

$$\cos\theta_x = \frac{x}{|\mathbf{a}|}$$

$$\cos\theta_y = \frac{y}{|\mathbf{a}|}$$

$$\cos\theta_z = \frac{z}{|\mathbf{a}|}$$

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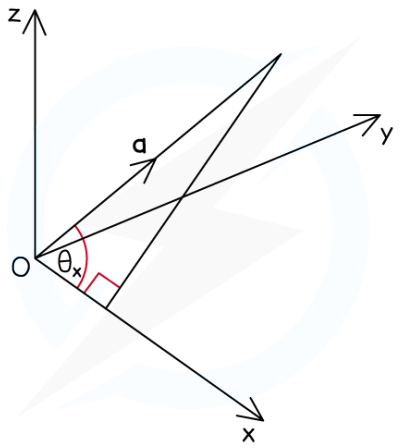
THESE FORMULAE COME FROM APPLYING SOHCAHTOA (SPECIFICALLY CAH) TO A RIGHT–TRIANGLE FORMED BY THE VECTOR AND ONE OF THE COORDINATE AXES.



IN THE DIAGRAM ABOVE FOR EXAMPLE, THE LENGTH OF THE SIDE ADJACENT TO ANGLE θ_x IS SIMPLY THE **x** COMPONENT OF THE VECTOR $\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, WHILE THE LENGTH OF THE HYPOTENUSE IS $|\mathbf{a}|$.

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- These formulae are not in the exam formulae book
- However, they may be derived using SOHCAHTOA as shown in the diagram
- The formulae also work for vectors in two dimensions

Examiner Tips and Tricks

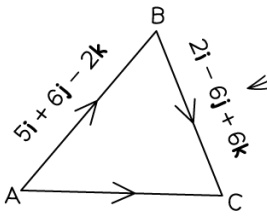
- If there is a diagram, labelling all known vectors and quantities will help, and don't worry about trying to make your diagram 3-D, as long as you label it well.

Your notes

Worked Example

The points A, B and C are such that $\overrightarrow{AB} = 5\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}$ and $\overrightarrow{BC} = 2\mathbf{i} - 6\mathbf{j} + 6\mathbf{k}$.

Show that ABC is an isosceles triangle.



DRAW A DIAGRAM

TO SHOW ABC IS AN ISOSCELES TRIANGLE WE MUST SHOW TWO SIDE LENGHTS ARE EQUAL

FIND ANY MISSING VECTORS

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$$
$$= (5\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}) + (2\mathbf{i} - 6\mathbf{j} + 6\mathbf{k})$$
$$= 7\mathbf{i} + 4\mathbf{k}$$

$6\mathbf{j} - 6\mathbf{j} = 0\mathbf{j}$ SO DOES NOT APPEAR

FIND MAGNITUDE OF EACH SIDE

$$|\overrightarrow{AB}| = \sqrt{5^2 + 6^2 + 2^2} = \sqrt{65}$$

$$|\overrightarrow{BC}| = \sqrt{2^2 + 6^2 + 6^2} = \sqrt{76}$$

$$|\overrightarrow{AC}| = \sqrt{7^2 + 4^2} = \sqrt{65}$$

TWO SIDES ARE EQUAL LENGTH

$|\overrightarrow{AB}| = |\overrightarrow{AC}|$ THEREFORE TRIANGLE ABC IS ISOSCELES

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